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THE ECONOMICS OF TRADEMARK LAW*

By William M. Landes** and Richard A. Posner***

That the law of intellectual property, including trademark law, can be analyzed in economic terms is no longer an insight with any power to astonish or even to offend.¹ Still needed, however, is an analysis that formalizes the economics of trademarks, relates trademarks to other forms of property, brings to bear the nascent economics of language and communication, and discusses and interrelates the principal doctrines of trademark law.² To supply this need is the task we have set ourselves in this article. It is an essay in positive rather than normative "law and economics." We use economics to explain the structure of trademark law rather than to urge changes in that law. The overall conclusion is that trademark law, like tort law in general (and trademark law is part of the branch of tort law known as unfair competition), can best be explained on the hypothesis that the law is trying to promote economic efficiency.³

I. THE ECONOMIC THEORY OF PROPERTY

The analysis draws heavily on the economics of property rights in law.⁴ A property right is a legally enforceable power to exclude others from using a resource, without need to contract with them. A property right confers two types of economic benefit,

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1. Previous economic analyses of trademark law include Ralph H. Folsom and Larry L. Teply, *Trademarked Generic Words*, 89 *Yale Law Journal* 1323, 70 *TMR* 206 (1980); Peter E. Mims, *Promotional Goods and the Functionality Doctrine: An Economic Model of Trademarks*, 63 *Texas Law Review* 639 (December 1984); Lee R. Burgunder, *An Economic Approach to Trademark Genericism*, 23 *American Business Law Journal* 391 (Fall 1985); Nicholas Economides, *The Economics of Trademarks* (Working Paper No. 21, Columbia University, Center for Law and Economics Studies, 1986); Lars H. Liebler, *Trademark Law, Economics and Grey-Market Policy*, 62 *Indiana Law Journal* 753 (Summer 1986-1987). Two recent judicial decisions apply economics explicitly to trademark law: *Scandia Down Corp. v. Euroquilt, Inc.*, 772 F.2d 1423, 227 USPQ 138 (CA 7 1985) (Easterbrook, J.); *W.T. Rogers Co. v. Keene*, 778 F.2d 334, 228 USPQ 145 (CA 7 1985) (Posner, J.). For a lucid summary of the basic doctrines of trademark law, see Arthur R. Miller and Michael H. Davis, *Intellectual Property: Patents, Trademarks, and Copyrights: In a Nutshell* 155-231 (1983); for excellent fuller treatments see J. Thomas McCarthy, *Trademarks and Unfair Competition* (2d ed 1984); Jerome Gilson, *Trademark Protection and Practice* (1974).

2. The articles cited in *supra* note 1 each focus on just one trademark doctrine or trademark problem.

3. See William M. Landes and Richard A. Posner, *The Economic Structure of Tort Law* (1987).

4. See Richard A. Posner, *Economic Analysis of Law*, Parts 1 and 2 (3d ed 1986).

static and dynamic. The former is illustrated by a natural (that is, uncultivated) pasture. If the owner cannot exclude others from using his pasture, there will be overgrazing; users of the pasture will ignore the costs they impose on each other in reducing the cattle's weight by making the cattle expend more energy in grazing in order to find enough to eat. The dynamic benefit of a property right is the incentive that the right imparts to invest in the creation or improvement of a resource in period 1 (for example, planting a crop) if no one else can appropriate the resource in period 2 (harvest time). For example, a firm is less likely to expend resources on developing a new product if competing firms that have not borne the expense of development can duplicate the product and produce it at the same marginal cost as the innovator; competition will drive price down to marginal cost, and the sunk costs of invention will not be recouped.⁵

Property rights impose costs as well as confer benefits. The costs of property rights are fourfold. The first is the cost of transferring such rights. If it is too high, a property right may prevent optimal adjustments to changing values. Suppose a factory is assigned a property right to the use of a river that runs beside it, because the river is more valuable as a sewer than for recreation, but as the years go by the relative values of these uses reverse. If the recreational users are numerous, the transaction costs of their buying the right to use the river from the factory may exceed the value of the right to them. In such a case, a liability rule would be better. The factory could be induced to discontinue its use of the river by being made to pay damages equal to the costs of the pollution to recreational users. The rule would reallocate the use of the river in accordance with changed values, without requiring a transaction.

The second major cost of a property rights system is what economists call "rent seeking." Suppose a ship has sunk and it has a salvage value of one million dollars, while the cost of salvage is only one-hundred thousand dollars. The potential gain (a form of economic rent) to the salvor is nine-hundred thousand dollars if a property right in the sunken ship can be acquired—and the competition to acquire it may eat up all or most of the potential

5. Both the static and the dynamic benefits of property rights presuppose that there are too many potential users of the property for transactions with all of them to be economical. When transaction costs (which in general, though not always, are a positive function of the number of contracting parties) are low, the Coase Theorem implies that enforceable contract rights are all that are needed to achieve optimal use and investment. See Ronald H. Coase, *The Problem of Social Cost*, 3 *Journal of Law and Economics* 1 (October 1960).

rents, transforming them into social costs.⁶ Intellectual property sometimes creates serious problems of rent seeking, because the resource is continuously created or discovered rather than being already owned. Like the sunken ship whose owner has abandoned it, it is waiting to be invented.

The third cost of property rights is that of protection and enforcement. It includes the costs incurred by police and courts in preventing trespass and theft, as well as the cost of a fence used to mark boundary lines or the cost of a registry used to record land titles. Intellectual property is particularly costly to protect. An idea cannot be seen, the way a piece of land can be. The land might have been transferred by inheritance for many generations but it is the same piece of land, recorded in the same land registry. Tracing the descent of an idea is harder. Moreover, because of the "public good" character (roughly, the ease of using without paying) of intellectual property, it can be costly, without legal protection, to prevent misappropriation and exclude free riders.

The last cost of property rights is the social cost of restricting the use of property when it has a public good character. In the case of farmland, whether cultivated or uncultivated, adding a user will impose costs on the existing user(s); so the fact that a fence keeps additional users out need not impose a net cost. But often, adding users will not impose costs on previous users of intellectual property—not directly anyway.⁷ One farmer's using the idea of crop rotation does not prevent other farmers from using the same idea.⁸ When the marginal cost of using a resource is zero, excluding someone from using it creates a deadweight loss, in addition to the cost of enforcing exclusion.

Since intellectual property is a particularly costly form of property, we would expect (and we find) that it is limited in ways that physical property is not.⁹ For example, the requirement that an invention, to be patentable, not be obvious excludes property rights in inventions when excessive rent seeking would be a serious problem if such rights were recognized. For "obviousness" implies

6. This example assumes, of course, that the original owner of the ship abandoned it, so that it is unowned. If it has not been abandoned, the owner can simply auction off the right to salvage the ship to the highest bidder, and there will be no rent-seeking problem.

7. It may of course discourage investment by preventing the previous users from recouping their sunk costs. This is the familiar "access v. incentives" trade-off discussed in many intellectual property cases.

8. Of course, when more farmers use crop rotation, output will rise and price will fall. Hence, extending crop rotation to other farmers imposes pecuniary costs on farmers already using crop rotation. We ignore these effects.

9. See, for example, Posner, *supra* note 4 at 36-37; Edmund W. Kitch, *Graham v. John Deere Co.*; *New Standards for Patents*, 1966 *Supreme Court Review* 293 (1966), 49 *JPOS* 237 (April 1967).

a low cost of discovery and development and therefore a large potential gap between value and cost—a large opportunity to obtain economic rents. The limited duration of patents limits rent seeking by putting a ceiling on the expected value of the patent. It also reflects the high cost of tracing an idea over a long period of time in which it may have become embodied in a great variety of inventions.

II. THE ECONOMICS OF TRADEMARKS

A. Introduction

To oversimplify somewhat, a trademark is a word, symbol, or other signifier used to distinguish a good or service produced by one firm from the goods or services of other firms. Thus SANKA designates a decaffeinated coffee made by General Foods Corp.; XEROX the dry copiers made by Xerox Corporation; BIB—the MICHELIN MAN—tires made by the Michelin Company. A stylized penguin is the symbol of a line of paperback books published by Penguin Books; a distinctively shaped green bottle is a trademark of the producer of Perrier bottled water; the color pink is a trademark for residential insulation manufactured by Owens-Corning.

1. Benefits of Trademarks

Suppose you like decaffeinated coffee made by General Foods. If General Foods' brand had no name, to order it in a restaurant or grocery store you would have to ask for "the decaffeinated coffee made by General Foods." This would take longer to say, require you to remember more, and require the waiter or clerk to read and remember more than if you could just ask for SANKA. The problem would be even more serious if General Foods made more than one brand of decaffeinated coffee—as in fact it does. The benefit of a brand name is analogous to that of designating individuals by last as well as first names, so that, instead of having to say "the Geoffrey who teaches constitutional law at the University of Chicago Law School—not the one who teaches corporations," you can say "Geoffrey Stone—not Geoffrey Miller." In economic terms, the brand name reduces "consumer search costs."

To perform this economizing function, a trademark or brand name (these are rough synonyms) must not be duplicated. For another maker of decaffeinated coffee to sell its coffee under the name "Sanka" would destroy the benefit of the name in identifying a brand of decaffeinated coffee made by General Foods. (Whether

there might be offsetting benefits is considered later.) It would be like allowing a second rancher to graze his cattle on a pasture the optimal use of which required that only one herd be allowed to graze. Moreover, as we are about to see, the failure to enforce trademarks would impose two distinct costs—one in the market for trademarked goods and the other in the market in language.

(a) The Market for Trademarked Goods

The benefits of trademarks in reducing consumer search costs require that the producer of a trademarked good maintain a consistent quality over time and across consumers. Hence trademark protection encourages expenditures on quality. Suppose a consumer has a favorable experience with brand X and wants to buy it again. Or suppose he wants to buy brand X because it has been recommended by a reliable source or because he has had a favorable experience with brand Y, another brand produced by the same producer. Rather than investigating the attributes of all goods in order to determine which one is brand X or is equivalent to X, the consumer may find it less costly to search by identifying the relevant trademark and purchasing the corresponding brand. For this strategy to be efficient, however, not only must it be cheaper for the consumer to search for the right trademark than for the desired attributes of the good, but past experience must be a good predictor of the likely outcome of current consumption choices. That is, the brand must exhibit consistent quality. Thus, a trademark conveys information that allows the consumer to say to himself, "I need not investigate the attributes of the brand I am about to purchase because the trademark is a shorthand way of telling me that the attributes are the same as that of the brand I enjoyed earlier."¹⁰

Less obviously, a firm's incentive to invest resources in developing and maintaining (as through advertising) a strong mark depends on its ability to maintain consistent product quality. In other words, trademarks have a self-enforcing feature. They are valuable because they denote consistent quality, and a firm has an incentive to develop a trademark only if it is able to maintain consistent quality. If a brand's quality is inconsistent, consumers will learn that the trademark does not enable them to relate their

10. As emphasized in John F. Coverdale, *Trademarks and Generic Words: An Effect-on-Competition Test*, 51 *University of Chicago Law Review* 868 (Summer, 1984), the benefits of a trademark to the consumer need not depend on whether the trademark identifies a particular brand or identifies the producer of that brand. Consumers benefit even if they are unable to identify the producer of a brand they desire to purchase. See *Bayer Co. v. United Drug Co.*, 272 Fed 505, 509 (SDNY 1921) (L. Hand, J.).

past to future consumption experiences. The branded product will then be like a good without a trademark; the trademark will not lower search costs, so consumers will be unwilling to pay more for the branded than for the unbranded good. As a result, the firm will not earn a sufficient return on its expenditures promoting the trademark to justify the expenditures. A similar argument shows that a firm with a valuable trademark would be reluctant to lower the quality of its brand because it would suffer a capital loss on its investment in the trademark.¹¹

It should be apparent that the benefits of trademarks in lowering consumer search costs presuppose legal protection of trademarks. The value of a trademark is the saving in search costs made possible by the information or reputation that the trademark conveys or embodies about the brand (or the firm that produces the brand). Creating such a reputation requires expenditures on product quality, service, advertising, and so on. Once the reputation is created, the firm will obtain greater profits because repeat purchases and word-of-mouth references will generate higher sales, and because consumers will be willing to pay higher prices in exchange for lower search costs and greater assurance of consistent quality. However, the cost of duplicating someone else's trademark is small—the cost of duplicating a label, design, or package where the required inputs are widely available—and the incentive to incur this cost (in the absence of legal regulation) will be greater the stronger the trademark. The free-riding competitor will, at little cost, capture some of the profits associated with a strong trademark because some consumers will assume (at least for a time) that the free rider's brand and the original trademark holder's brand are identical. If the law does not prevent it, free riding will eventually destroy the information capital embodied in a trademark. The prospect of free riding may therefore eliminate the incentive to develop a valuable trademark in the first place.

(b) The Market in Languages

An entirely different benefit of trademark protection derives from the incentives that such protection creates to invest resources

11. See Benjamin Klein and Keith B. Leffler, *The Role of Market Forces in Assuring Contractual Performance*, 89 *Journal of Political Economy* 615 (1981); Carl Shapiro, *Premiums for High Quality Products as Returns to Reputations*, 98 *Quarterly Journal of Economics* 659 (1983).

not in maintaining quality but in inventing new words¹² (or symbols or, less clearly, design features used as trademarks, such as the Perrier bottle—but for the moment we confine our attention to words). Trademarks improve the language in three ways. They increase the stock of names of things, thus economizing on communication and information costs in the ways just suggested. They create new generic words—words that denote entire products, not just individual brands. (“Aspirin,” “brassiere,” “cellophane,” “thermos,” “yo-yo,” “dry ice,” and a number of other names of common products were once trademarks in the United States¹³—and, whatever courts might say, KLEENEX and XEROX are widely used to denote entire products as well as particular brands.)¹⁴ And they enrich the language, by creating words or phrases that people value for their intrinsic pleasingness as well as their information value (PHEREMON perfume, SWAN’S DOWN cake mix).

These benefits, however, are small. (This point will later help explain important features of trademark law—such as the termination of trademark protection if the mark becomes generic—that would be inexplicable if trademarks provided the same sort of intellectual enrichment that patents and copyrights do.) To show this will require a brief excursion into the economics of language.¹⁵ The goal (to which language is central) of a communication system is to minimize the sum of the costs of avoiding misunderstanding and the costs of communicating. Suppose we have a word for snow and a word for falling, and now the question is, should there be a new word meaning “falling snow”? In favor of the new word is the fact that unless it is very long it will be shorter to speak, read, and write; against it is that people will have to learn and remember another word. The more common a

12. A linguistic study of two-thousand brand names concludes that they are formed on the same linguistic principles as other words. See Jean Praniskas, *Trade Name Creation: Processes and Patterns* 101 (1968). For a fascinating older study, reaching a similar conclusion, see Louise Pound, *Word-Coinage and Modern Trade Names*, 4 *Dialect Notes* 29 (1913).

13. Aspirin, *supra* note 10; Brassiere: *Charles R. DeBevoise Co. v. H. & W. Co.*, 69 NJ Eq 114, 60 A 407 (1905); Cellophane: *DuPont Cellophane Co. v. Waxed Products Co.*, 85 F2d 75, 30 USPQ 332 (CA 2 1936); Thermos: *King-Seeley Thermos Co. v. Aladdin Industries, Inc.*, 321 F2d 577, 138 USPQ 349 (CA 2 1963); Yo-Yo: *Donald F. Duncan, Inc. v. Royal Tops Mfg. Co.*, 343 F2d 655, 144 USPQ 617 (CA 7 1965); and Dry Ice: *Dry Ice Corp. of America v. Louisiana Dry Ice Corp.*, 54 F2d 882, 12 USPQ 199 (CA 5 1932).

14. Adrian Akmajian, Richard A. Demers and Robert M. Harnish, *Linguistics: An Introduction to Language and Communication* 70 (1984), give these two words as examples of how brand names have become part of ordinary language.

15. This is an undeveloped branch of economics. The only previous economic analysis of language that we have found is a short paper by Jacob Marschack, *Economics of Language*, 10 *Behavioral Science* 135 (1965). A related subject, the economics of rhetoric, is discussed briefly in Richard A. Posner, *The Economics of Justice* 172-73, 276-77, 280, 282 (1981) [hereinafter cited as Posner-Economics].

term is, the more the benefits of having a single word are likely to outweigh the costs, not only because the gains from shortening the term will be greater, but also because the cost of learning and remembering a word is less if it is in common use. So we are not surprised to find that Eskimos have a single word for falling snow, but we do not.¹⁶ The use of a word rather than a periphrasis to name a brand illustrates the same point.

Both examples are closely related to a statistical observation made years ago: the length of words is inverse to their frequency.¹⁷ It might seem that, rather than frequently used words being shorter than infrequently used words, all words would be short in order to economize on communication costs. But length is an important dimension along which words vary (and can thus be distinguished from one another), and this dimension would be lost if all words were short. It makes economic sense for the frequently used words to be short and the infrequently used ones to be long; then total length is minimized without sacrificing distinctiveness, which would increase the number of errors (misunderstanding). More generally, the drive to make language simple is balanced by the desire to avoid ambiguities and confusions that result from lack of differentiation.¹⁸

Here are some other examples of efficient language rules.

1. Irregularities of grammar and spelling are more common in frequent than infrequent words.¹⁹ The more frequently used a word is, the easier it is to learn by rote, and hence the less important it is that people be able to construct the word by the application of a rule. Everyone knows that the past participle of the verb "to be" is "been"; but it is convenient to be able to construct the past participle of "excogitate" by rule rather than have to memorize it.

2. Pronunciation changes faster than spelling because changes in pronunciation do not reduce the intelligibility of existing reading matter, which represents a vast and valuable capital stock of knowledge.

3. Perfect synonyms are rare; they would increase learning costs without adding to the communication resources of the language (except that synonyms make it easier to write poetry that rhymes or has regular meter).

16. See F.R. Palmer, *Semantics* 21 (2d ed 1981).

17. See George Kingsley Zipf, *Human Behavior and the Principle of Least Effort* 24 (1949), discussed in Marschack, *supra* note 15 at 139-40.

18. See Jean Aitchison, *Language Change: Progress or Decay* 201, 226 (1981).

19. See, for example, Theodora Bynon, *Historical Linguistics* 43 (1977).

4. Pronouns, which in all known languages are short, are an ingenious device for economizing on the length of words.

Examples of the efficiency of language rules could be multiplied,²⁰ but the most important thing for present purposes is to note that efficiency is achieved without a system of property rights in words, in grammatical forms, and so on. Of course, the costs of enforcement—the costs, for example, of a system under which the coiner of a word (such as Jeremy Bentham, who coined “codification,” “minimize,” and several other words still in common use) obtained a property right in it—would be staggering. This may be a sufficient explanation for why there is no such system. Yet it seems (though this is just a guess) that even without property rights, the language has attained a reasonable degree of efficiency. Of particular relevance to trademarks is the fact that the creation of new words for new things seems not to be retarded by the fact that the coiner of a word can obtain no property right (no “trademark,” so to speak). Either the costs of thinking up new words are slight, or the incentives to do so, independent of any direct compensation, are great. The former seems the case with proper names (naming a baby, for example) and for terms of art. The latter seems the case with trademarks. If a producer wants to market a new brand effectively, he needs a distinctive name—unless of course he is trying to pass off his brand as someone else’s.

This analysis suggests that we do not need trademark protection just to be sure of having enough words, though we may need patent protection to be sure of having enough inventions, or copyright protection to be sure of having enough books, movies, and musical compositions. It is true that computer operating systems, which are a form of language, are copyrightable; and maybe an invented language, such as Esperanto, would be. But the investment required to create a new language is greater than that required to create a new word, so the case for property rights is stronger in the former instance than in the latter.

2. The Costs of Legally Enforceable Trademarks

These costs are modest, at least in the simple case of the “fanciful” mark, such as EXXON and KODAK, which has no information content except to denote a specific producer or brand. Since the mark “goes with” the brand (in a sense explained later), the transfer of the mark is automatically effected by a transfer of

20. See, for example, Aitchison, *supra* note 18 at 152-55 and ch 8.

the rights to make the branded product, as by selling or licensing the production rights or assets. Unlike the case of the sunken ship, rent seeking to stake out a trademark is not much of a problem either. The distinctive yet pronounceable combinations of letters to form words that will serve as a suitable trademark are as a practical matter infinite, implying a high degree of substitutability and hence a slight value in exchange. Finally, the costs of enforcement, though not trivial (especially where there is a danger of a brand name's becoming a generic name), are modest, and (again putting aside the generic problem) do not include the cost in inefficient resource allocation from driving a wedge between price and marginal cost. A proper trademark is not a public good; it has social value only when used to designate a single brand.

We may seem to be ignoring the possibility that, by fostering product differentiation, trademarks may create deadweight costs, whether of monopoly or (excessive) competition. We have assumed that a trademark induces its owner to invest in maintaining uniform product quality; but another interpretation is that it induces the owner to spend money on creating, through advertising and promotion, a spurious image of high quality that enables monopoly rents to be obtained by deflecting consumers from lower-price substitutes of equal or even higher quality. In the case of products that are produced according to an identical formula, such as aspirin or household liquid bleach, the ability of name-brand goods (Bayer aspirin, Clorox bleach) to command higher prices than generic (nonbranded) goods has seemed to some economists and lawyers an example of the power of brand advertising to bamboozle the public and thereby promote monopoly;²¹ and brand advertising presupposes trademarks—they are what enable a producer to readily identify his brand to the consumer. Besides the possibility of creating monopoly rents, trademarks may transform rents into costs as one firm's expenditure on promoting its mark cancels out that of another firm. Although no monopoly profits are created, consumers may pay higher prices, and resources may be wasted in a sterile competition.

The short answer to these arguments is that they have gained no foothold at all in trademark law, as distinct from antitrust law. The implicit economic model of trademarks that is used in that

21. See, for example, *Borden, Inc. v. FTC*, 674 F2d 498 (CA 6 1982) (the ReaLemon case); William S. Comanor and Thomas S. Wilson, *Advertising and Market Power*, ch 3 (1974); Richard Schmalensee, *On the Use of Economic Models in Antitrust: The ReaLemon Case*, 127 *University of Pennsylvania Law Review* 994 (1979); Warren G. Lavey, *Patents, Copyrights, and Trademarks as Sources of Market Power in Antitrust Cases*, 27 *Antitrust Bulletin* 433, 448-51 (Summer 1982).

law is our model, in which trademarks lower search costs and foster quality control rather than promote social waste and consumer deception. A longer answer, which we shall merely sketch, is that the hostile view of brand advertising has been largely and we think correctly rejected by economics.²² The fact that two goods have the same chemical formula does not make them of equal quality to even the most coolly rational consumer. That consumer will be interested not in the formula but in the manufactured product and may therefore be willing to pay a premium for greater assurance that the good will actually be manufactured to the specifications of the formula. Trademarks enable the consumer to economize on a real cost because he spends less time searching to get the quality he wants. If this analysis is correct, the rejection by trademark law of a monopoly theory of trademarks is actually a mark in favor of the economic rationality of that law.

B. A Model

The essential economic function of trademarks—that of reducing consumer search costs—can be given greater precision and rigor in a simple model.²³ The key analytic move, and heart of our analysis, is to define the “full price” of a good as its money price plus the search costs incurred by the buyer in obtaining information about the relevant attributes of the good. If search costs decline because consumers have more information, the producer may be able to raise his price without exceeding the full price that the consumer is willing to pay. For example, if a brand’s money price is eight dollars, and consumer search costs are two dollars, making a full price of ten dollars, and search costs fall to one dollar, the producer can raise his price by one dollar without the consumer’s feeling worse off. A firm produces information that reduces consumer search costs in part through its trademark. The more resources the firm spends developing and promoting its mark, the stronger will its mark be and the lower, therefore, consumer search costs will be; so the firm will be able to charge a higher price.

The information generated by a trademark is of two sorts. One is information that enables the consumer to identify the

22. See Klein and Leffler, *supra* note 11, and references there.

23. The mathematical version of this model appears in Landes and Posner, *supra* note 3 at 275-80. The model is similar to that developed in Isaac Ehrlich and Lawrence Fisher, *The Derived Demand for Advertising: A Theoretical and Empirical Investigation*, 72 *American Economic Review* 366 (1982), which in turn builds on the approach to advertising in Gary S. Becker and George J. Stigler, *De Gustibus Non Est Disputandum*, 67 *American Economic Review* 76 (1977). Ehrlich and Fisher treat advertising as a variable that reduces search costs and thus the full price of the good.

source of the good—for example, knowing that CREST toothpaste comes from a single source even if one does not know that Procter & Gamble Company is that source. Information about source economizes on search costs by lowering the cost of selecting goods on the basis of past experience or the recommendation of other consumers. Imagine how high search costs would be if all producers of toothpaste used the identical mark or no mark at all. The other sort of information produced by the trademark is information about the product itself. For example, a “descriptive” mark (of which more later) may, in addition to identifying source, describe some properties of the brand; this information also lowers search costs.

Consumer search costs also depend on the availability of words (or other symbols) that a firm can use as its trademark. Most of the time, as was said, the number of available words will be too large to affect the analysis, but if one firm is given exclusive rights to words that define or describe a product, this will reduce the productivity of the trademarks of other firms that make the same product and thus will increase the search costs of those firms’ consumers. Suppose, for example, that a particular manufacturer of personal computers could not use the terms “personal computer” or “PC” to advertise its computer because another manufacturer had the exclusive right to these terms. It might therefore have to describe its product as “a machine capable of doing word processing and high speed calculations using a central processing unit,” or in some other prolix and roundabout way. (An even more elaborate periphrasis would be necessary if “word processing” had also been appropriated.) Because it is harder to recall long than short phrases, a lengthy advertising message that is difficult to recall will convey less information about the firm’s product, so search costs will rise.

Since a good’s full price is the sum of its money price and the consumer’s search costs, which are lower the stronger the seller’s trademark is, a seller will have an incentive to expend resources on strengthening its trademark; for this expenditure, by reducing the consumer’s search costs, will enable the seller to raise the (money) price for its good. As a result, even if the seller’s market is competitive—no seller has market power or monopoly profits—different brands of the “same” product may sell at different prices. Firms with strong trademarks will command higher prices for their brands than other firms in the market, not because of any market power, but because the search costs associated with their brand are lower. For example, if competition constrains the

full price of the product to two dollars, a firm whose brand costs the consumer one dollar in search costs will sell its brand for one dollar, while a firm whose brand costs the consumer fifty cents in search costs will sell its brand for one dollar fifty cents, even if the two brands are (physically) identical.

The assumption that all firms produce brands of identical quality is useful in showing that prices may differ among brands not because of any quality differences in the underlying physical product but because of differences in the strength of the brands' trademarks. Yet an important and widely recognized benefit of trademarks is that they give firms an incentive to improve the quality of their products. Without an exclusive right to use one's own trademark, a firm that was producing a lower-quality brand could hope, by duplicating the trademark of a superior brand, to mislead consumers into believing that the brands were equivalent. By making it more costly for consumers to distinguish higher-quality from lower-quality brands, this duplication would reduce a firm's incentive to incur the added cost that would be necessary to produce a higher quality. Hence the average quality of the product as a whole would be lower than with legally enforceable trademarks.

III. THE ECONOMICS OF SPECIFIC TRADEMARK DOCTRINES

A. The Acquisition, Transfer and Duration of Trademarks

1. How Trademarks Are Acquired

One of the costs of a property rights system—the transformation of the rents flowing from possession of a valuable right into costs of acquiring the right in the first place—is a potential problem with trademarks. There are three systems in use for regulating the acquisition of trademarks. One is registration. It resembles the systems used for acquiring patents and copyrights and is in use in most of the world outside the United States. The second system, which is the traditional approach of Anglo-American common law, is a type of "first possession" rule and is thus analogous to the common law method of acquiring property rights in wild animals, oil and gas, and certain other resources (including, in some states, water). The third system, the current American system, is a mixture of registration and first possession.

Under the common law approach, the possession that confers ownership of a trademark is defined as use in commerce, which

means sale to intended customers (typically, the public).²⁴ There are several advantages of a first-possession rule.

(a) It minimizes rent seeking. A firm allowed to register trademarks without using them might invest substantial resources in thinking up plausible new brand names. For even if, as we have suggested, the availability of such names is very great, the ownership of a vast number of them, and the aggregate licensing revenues that such ownership would command, would be a magnet drawing resources into the activity of creating brand names, probably beyond the optimal level of such investment. Apparently the "banking" of trademarks in countries such as Japan that have a pure registration system does occur and has made it more costly to enter markets in those countries.

(b) First possession reduces administrative costs compared to a system that requires deciding who first invented the trademark. Since trademarks often consist of common words, shapes, or colors, it would be costly to figure out which party to a trademark dispute had invented the mark first. A cheaper (not costless) alternative is to determine who used it first. An even cheaper alternative is who registered it first, but it leads to the problem discussed in the preceding paragraph.

(c) Use in commerce means sale of the good, with the trademark attached, to the public. Thus, a potential second comer will be on notice not to invest resources in developing a mark similar or identical to one already in use. Potential duplication is not avoided completely because, as we shall see in a moment, there will be an interval between developing and fully exploiting the mark during which another person (the "junior user") may be developing the same mark unaware of the first ("senior user").²⁵ But the potential duplication costs probably are lower, on the average at least, than in the case of patents, because the cost of adopting a new trademark is normally less than that of inventing a new product or process.²⁶

(d) Basing the property right on use fits in with the social function of trademarks in identifying and distinguishing goods. If the trademarked good is not available for sale, the trademark confers no benefit. Thus, conditioning trademark rights on use is

24. See, for example, 1 McCarthy, *supra* note 1, §§16.2 and 16.3.

25. See *Blue Bell, Inc. v. Farah Mfg. Co.*, 508 F2d 1260, 185 USPQ 1 (CA 5 1975).

26. The function of the patent system in reducing duplication by warning off prospective inventors of the same product or process is emphasized in Edmund W. Kitch, *The Nature and Function of the Patent System*, 20 *Journal of Law and Economics* 265 (October 1977). This is an example of how the law of intellectual property tries to control the problem of rent seeking.

a way of limiting the use of scarce enforcement resources to situations in which the rights in question are likely to yield net social benefits. The solution is not ideal; it could lead to the premature development and marketing of goods by a firm eager to establish a right in a nifty trademark. But if the supply of brand names is as copious as we believe, very few individual trademarks will be so valuable apart from the products that they name that a firm will distort its marketing decisions in order to appropriate a particular name.

The current American system of establishing trademark rights is a mixture of state common law rights and an optional federal registration system (itself a mixture of registration and first-possession principles) under the Lanham Act.²⁷ Registration under the Lanham Act does not confer a property right without use, but less use is required than at common law; often a token sale or single shipment will suffice.²⁸ The main social benefit of a federal registration system is that notice is likely to be more widespread, so that inadvertent duplication is less likely; hence use becomes a less important method of preventing duplication. But the federal registration system involves significant costs. Like a pure registration system, it may enable firms to "bank" trademarks, if the use requirement really is minimal. If the sale of trademarks apart from the goods they denote is forbidden (which it is, as we shall see in the next section), "banking" trademarks may not only draw excessive resources into creating trademarks, but also force competitors of the trademarks' owner to adopt less efficient trademarks, since they could not buy them from the "bank."

The biggest problem with a first-possession rule for intellectual property, and the strongest argument therefore for a system of paper titles (a trademark registry—corresponding, as we have noted, to the patent registry), is that the thing possessed has no definite physical locus. Suppose that producer A, who makes brand X desk lamps, is at present selling only in New York State but has plans to sell eventually throughout the country. Can producer B, who operates only in California, sell the desk lamps under the X name, on the theory that A is not using the trademark in California? Or if A is selling a hammer throughout the country

27. Sections 1 et seq, 15 USC §§1051 et seq. The Act also provides remedies for infringement of unregistered (that is, common law) trademarks and for false advertising and labeling.

28. See, for example, *Blue Bell, Inc. v. Farah Mfg. Co.*, supra note 25 at 1266-67, 185 USPQ 1; *Maternally Yours, Inc. v. Your Maternity Shop, Inc.*, 234 F2d 538, 542, 110 USPQ 462 (CA 2 1956).

under the name X and plans eventually to sell a full line of tools under the name, can B affix the name to his own brand of screwdriver? If A eventually begins to sell desk lamps in California or eventually begins to produce screwdrivers as well as hammers, who owns the X trademark—A or B?

The courts resolve these issues in a way that may minimize the costs arising from duplication of trademarks, but is rather costly and uncertain in administration. To determine whether the goods sold by A and B (if and when A completes his plans of expansion, whether geographic or product as the case may be) will be too similar to share the same name without unduly confusing the consumer, and if so whether B should be deemed an infringer, the courts primarily consider the proximity or similarity of A's original and expansion uses, A's unreasonable delay, if any, in enforcing his trademark against B (i.e., A's laches), and B's good or bad faith—whether he knew about A's trademark and was copying it or whether it was a coincidence that he began using the same mark.²⁹ The closer A's original and expansion uses are, the costlier it will be for A and its customers if A is forced to use a different mark in the expansion uses. Given the mobility of consumers,³⁰ they will be confused to find that the same brand is called one thing in one state and another thing in another. That is why Standard Oil Company of New Jersey came up with a new mark, EXXON, to replace the ESSO, HUMBLE, STANDARD, and ENCO marks that it had used for the identical products in different states. Consumers may also be confused if complementary goods made by the same producer, such as a hammer and a screwdriver, are sold under different names. So if A is denied the use of his trademark in his expansion markets, geographic or product, he may like Standard Oil of New Jersey be forced to adopt a wholly new trademark, thus sacrificing some reputation capital associated with the original mark.

The doctrine of laches forces A to internalize B's cost of duplication. If A has reason to know that B is proceeding to develop a duplicative mark in ignorance of A's prior use, A must, on pain of not being able to use his mark in his expansion markets, warn B off. If, however, B, far from proceeding in ignorance of A's prior use, has deliberately copied that mark (bad faith), the costs of

29. See, for example, *Polaroid Corp. v. Polarad Electronics Corp.*, 287 F2d 492, 128 USPQ 411 (CA 2 1961) (Friendly, J.); *Dwinell-Wright Co. v. White House Milk Co.*, 132 F2d 822, 56 USPQ 120 (CA 2 1943) (L. Hand, J.); 2 McCarthy, *supra* note 1, chs 24, 26 and 31.

30. Well illustrated by *Park 'N Fly, Inc. v. Dollar Park and Fly, Inc.*, 782 F2d 1508, 1509, 228 USPQ 853 (CA 9 1986), where the parties provided services to airline passengers at different airports, but the pool of customers was the same. We discuss the determinants of the likelihood of confusion in greater detail later.

duplication are self-imposed, and he is entitled to less consideration.

The Lanham Act has eased the problem of geographic overlap, thanks to imaginative judicial interpretation. The courts have interpreted the act to eliminate any good-faith defense for a firm using a trademark listed in the federal registry on the same product.³¹ The idea is that, before beginning to use a mark, the firm should check the registry; if it finds that the mark is being used on the same product, it cannot later claim good faith when sued for infringement. If the products are different (the hammer and screwdriver example), the defense is not automatically extinguished, because the path of expansion into different though related products is inherently uncertain. The same can be said for geographic expansion, but there is a critical difference. Even if the firm that is using a trademark in one part of the country never expands to other parts, consumers are not fixed in one place, and in traveling around the country or in moving from one part of the country to another they may be confused if different brands of the same product are sold under the same name. They are apt to assume that every desk lamp sold under a particular brand name is the same brand, that is, is produced by the same producer. The courts have eliminated this source of confusion for registered marks.

2. The Sale of Trademarks

The law generally prohibits the sale of trademarks except as an incident to selling the right to produce the good that the mark identifies.³² This rule that trademarks cannot be appropriated "in gross" may seem puzzling. Why should Coca-Cola Company, for example, not be allowed to sell its trademark while retaining all rights to its secret formula for syrup? The answer relates to the economic function of trademarks in providing useful information, which would be more costly to obtain elsewhere about a good's attributes. If A sells just his trademark to B and consumers know about the sale, attaching A's trademark to B's good will not, at

31. See 2 McCarthy, *supra* note 1, §26.13.

32. See, for example, *PepsiCo, Inc. v. Grapette Co.*, 416 F2d 285, 163 USPQ 193 (CA 8 1969); 1 McCarthy, *supra* note 1, §§18.1-18.2. A major exception is for "promotional goods," which are goods that carry the trademark but are not closely related to the goods primarily produced by the trademark owner. A good example is a tee shirt with the trademark or emblem of the Chicago Bears football team. This trademark has a value independent of the good it identifies. Thus, the trademark itself is a good. The analysis of trademarks does not deal with promotional goods; they are analyzed in Mims, *supra* note 1. Another major exception is the service mark (HOLIDAY INN, for example), often used to denote a franchise operation rather than a manufactured good. Here there is no manufacturer, so the sale of the trademark cannot be tied to the sale of a right to manufacture.

least as a first approximation, enable B to obtain a higher price for his good.³³ Contrast this with the case in which A sells the formula or other assets used to produce the good; here there is essentially just a change of ownership and hence no reason to believe that quality will be less. Where the only transfer is of the trademark itself, the knowledgeable consumer will be reluctant to pay more for the same old good (B's good) just because it has a new name. This implies that B will purchase A's mark in gross, for use on B's (unchanged) good, only if a sufficient number of consumers believe that the mark still identifies A's good (and hence are confused). In such a case, the expected value of A's trademark affixed to B's good will depend on the added price that B will be able to charge for his brand because consumers are confused. Recall from the model that trademarks lower search costs. If there is some probability that consumers believe that A's mark correctly identifies B's good, B will obtain a higher price for his (pseudo-A) good than he would without confusion. The increase in price will be higher the greater the likelihood of confusion and the stronger A's trademark is relative to B's.

This analysis is incomplete in three respects, however. First, it fails to consider that, if blocked from selling his trademark, A can accomplish the same end by reducing the quality of its own brand, and thus obtain the same profits from deceiving consumers as it could have obtained if allowed to sell its trademark to B for use on B's inferior good. Trademark law contains, however, a doctrine designed (perhaps not very effectively) to close this loop-hole. As McCarthy explains, "Since a trademark is not only a symbol of origin, but a symbol of a level of quality, a substantial change in the nature or quality of the goods sold under a mark may so change the nature of the thing symbolized that the mark becomes fraudulent or that the original rights are abandoned."³⁴

Second, the analysis ignores the market checks on A's selling his trademark for a deceptive use by B. Once consumers "wise up" to the fact that A's trademark does not denote a good of consistent quality, they will refuse to pay as much to either A or B. It might seem, therefore, that in selling his mark, as in licensing it, A would have an incentive to monitor the quality of B's good to make sure that B did not impair A's reputation by selling an inferior quality of good. Moreover, even if there were no doctrine that forbade the sale of trademarks in gross, A would forfeit the

33. One can think of exceptions. For example, if consumers believe that A's sale of its mark was contingent on its implicit certification of B's good, B may be able to obtain a higher price for its good.

34. 1 McCarthy, *supra* note 1, §17.9 at 784 (footnotes omitted).

trademark if a buyer attached the mark to an inferior good, for this is the rule when the trademark is sold lawfully, that is, along with the right to produce A's good.³⁵

But this ignores the fact that A, when he sells his trademark to B, may be leaving the market in which the trademarked good is sold or even going out of business entirely. In either of these "last-period" settings (as economists call them), A may face no prospect of market retaliation from selling a trademark that will be used to deceive consumers. An important aspect of the rule against sale in gross is, as a matter of fact, the law's unwillingness to allow a creditor to levy on a trademark of the bankrupt company. Unless the buyer of the bankrupt's estate continues the bankrupt's business, the bankrupt's trademarks are deemed abandoned.³⁶ In this setting, the rule acts as a prophylactic against the creditor's affixing the mark to an inferior good. It is prophylactic because, as noted earlier, the mark will be forfeited once it is discovered that it is being affixed to an inferior good, but that may take years to discover, and many consumers may be deceived in the interim.

As the analysis would predict, the rule forbidding assignments of trademarks in gross is, so far as we can determine, applied only in last-period cases. Although we have not made a complete search, the cases we have read in which a sale was voided as being in gross are all cases in which the seller was either no longer in business or no longer using the trademark himself.³⁷

Third, we have not explained B's incentive to try to gull consumers into thinking they are buying a superior brand. If consumers are easily gulled, our earlier rejection of the monopoly theory of product differentiation would be hard to sustain. If they are not easily gulled, B will suffer the same loss of reputation capital as A. In some cases, however, that prospect will not deter B. B may be in its last period; it may have little or no reputation capital to lose; or the cost of producing its inferior product may be so low that its short-run expected gain from deception exceeds its long-run loss of reputation capital.

3. The Duration of Trademarks

The lack of a fixed term for trademarks is one of the striking differences between trademarks, on the one hand, and copyrights

35. See id §18.8.

36. See id §18.9 at 818.

37. See, for example, *PepsiCo, Inc. v. Grapette Co.*, supra note 32; *Uncas Mfg. Co. v. Clark & Coombs Co.*, 309 F2d 818, 135 USPQ 282 (CA 1 1962); *American Photographic Publishing Co. v. Ziff-Davis Publishing Co.*, 135 F2d 569, 573, 57 USPQ 362 (CA 7 1943); 1 *McCarthy*, supra note 1, §18.5(D).

and patents, on the other. The difference, however, makes economic sense. If a given name has no scarcity value, so that it yields zero rents, perpetual compared to limited duration cannot create rent-seeking problems even if discount rates are very low (or zero or, for that matter, negative). And identification costs, which would plague perpetual patents, are not a serious problem either. The trademark is tied to physical property—the good that it names—and usually it is not costly to find out who the producer of a good, and therefore the trademark owner, is, though we shall consider some exceptions later. Moreover, to make the producer give up the name before he ceased selling the good would impose search costs on consumers; and when and if he ceases selling the good, the trademark lapses—another illustration of the doctrine of trademark abandonment.³⁸ We shall see, however, that trademarks are subject to variable limitations both of time and of “product space.”

B. The Requirement of Distinctiveness

1. Introduction

Trademark protection is available only for a word or other signifier that identifies the underlying good (or service) and distinguishes it from that of other producers. Lack of distinctiveness would make the mark incapable of identifying the good and recalling to a consumer the information (on the basis of previous experience with the good by him or other consumers) that lowers his search costs and by doing so enables the producer to charge a higher price. But even without a distinctive mark, a trademark may reduce search costs somewhat—so what would be wrong with trademark protection for nondistinctive signifiers? First, such protection might be unnecessary because no one would want to free ride on a nondistinctive signifier. The incentive to free ride depends on the difference between the profits generated by the mark—which by assumption are close to zero in the nondistinctive case—and the costs of duplication.

More important, since a mark that does not distinguish one brand from another must, virtually by definition, be using words, symbols, shapes, or colors similar to those used by other producers of X, the grant of legal protection to such a mark might prevent the other producers from continuing to use words that they require to compete effectively. Consumer search costs are a function not only of words used to denote a particular producer's brand but

38. On the doctrine generally, see 1 McCarthy, *supra* note 1, ch 17.

also of words used to denote the product, such as "computer" or "electrical." Since by definition they are words common to all or most brands, and since they are descriptive rather than arbitrary, their availability is limited. Letting a producer appropriate a non-distinctive mark would enable him to obtain, in effect, exclusive rights to such words, assuming that other producers—those with distinctive marks—would have to remove the words from their labels, packaging, or product design to avoid infringing the first producer's trademark, as they would have to do if their use of the words were deemed source-identifying and not merely descriptive. The result would be to raise the search costs of the consumers of those producers who no longer are permitted to use the descriptive words. Ultimately this would reduce the prices of their brands and lower the amount they produced. The industry supply curve would shift to the left, resulting in a social loss because consumers would be paying higher prices for a smaller quantity. The earlier example of a firm allowed to use "word processor" as a trademark illustrates this problem. The same problem would arise if producers were allowed to use the descriptive words in a nontrademark setting, but were required to add appropriate qualifiers and disclaimers to avoid infringing the trademark that employed those words. Those producers would incur added costs not borne by the owner of the trademark, and the result would be higher prices.

2. The Types of Trademark

The law could deal with the problem of a nondistinctive mark by requiring in every case an inquiry into the economic effects of allowing an exclusive right. The effect would be to make a trademark case very much like an antitrust case. Antitrust cases governed by the Rule of Reason are very costly to try (or even to settle), and the only thing that makes these costs (sometimes) worthwhile, both privately and socially, is the large private and social costs that some antitrust violations impose. Since the economic consequences of individual trademark abuses are pretty much limited to raising consumer search costs, the potential misallocations are much smaller than in most antitrust cases. It therefore would not pay, privately or socially, to conduct an antitrust-type analysis in most trademark cases. Instead, the law has classified potential marks by distinctiveness in a few broad categories and has made classification determinative of legality—much as in antitrust cases governed by *per se* rules. The result is sometimes

criticized for its crudeness,³⁹ but there are potentially offsetting reductions in administrative costs.

Let us consider the categories.⁴⁰ The so-called fanciful mark—the made-up name that resembles no other word, such as EXXON or KODAK—is the economically (and legally) least problematic. Much like a fanciful mark in their economic properties are arbitrary and suggestive marks. The former term refers to a word in common use that has no meaning related to the product that it is used to name; APPLE [Computer] and BLACK & WHITE [Scotch] are examples. The supply of such terms is very large. There are four-hundred fifty thousand words in Webster's Third New International Dictionary, and although they are not freely substitutable if one is trying to say something that will be understood, they are freely substitutable if one is uninterested in meaning.

Somewhat more problematic are suggestive marks—words that imply characteristics of the goods they are used to name but do not describe them. A good example is BUSINESS WEEK. The supply of suggestive marks is smaller, but not much smaller when one considers substitution between trademark categories. BUSINESS WEEK competes with FORBES and BARRON'S (arbitrary marks) as well as with THE WALL STREET JOURNAL—the last also an example of a suggestive mark.

Next in decreasing order of substitutability is the descriptive mark, such as ALL-BRAN or HOLIDAY INN. Here trademark protection is allowed only on proof of "secondary meaning," which means proof that the consuming public understands the word or phrase to name the brand.⁴¹ A given product has only so many attributes that interest buyers. So if one producer is allowed to appropriate the word that describes that attribute, he will obtain rents measured by the higher price he receives for his brand because it is so costly for his rivals to inform their customers of the attributes of their brands without using the descriptive word that has been appropriated. In time, however, the dictionary meaning of a descriptive word may go out of common use and the word may come to signify for most people just the name of the brand; ALL-BRAN has come to mean not any all-bran cereal but a particular brand of ALL-BRAN cereal. Once this happens, allowing

39. See, for example, *Scandia Down Corp. v. Euroquilt, Inc.*, supra note 1 at 1431 fn 3, 227 USPQ at 143.

40. Succinctly summarized in *Abercrombie & Fitch Co. v. Hunting World, Inc.*, 537 F2d 4, 9-11, 189 USPQ 759 (CA 2 1976), on rehearing 189 USPQ 769 (CA 2 1976).

41. See, for example, *J.M. Huber Corp. v. Lowery Wellheads, Inc.*, 778 F2d 1467, 228 USPQ 206 (CA 10 1985).

the word or expression to be appropriated may generate consumer benefits, by reducing confusion and search costs, that exceed the costs to rivals of being forbidden to use the same word.

Just as words can be classified into different types of trademarks, so can shapes and other signifiers. Similar to fanciful and arbitrary words are unusual symbols and shapes and combinations of well-known symbols, shapes, and colors. No problem arises when a firm appropriates such a signifier as its trademark. They are distinctive, so there is no question whether they are capable of providing source information; and their supply is virtually unlimited, so a competitor is not at a cost disadvantage in choosing another signifier as a trademark. Closely analogous to descriptive marks are common symbols (circles, squares, hearts, and the like) and individual colors (particularly primary colors). To allow a firm to appropriate one of these potential signifiers as its trademark creates the danger that, after several firms do this, the limited number of attractive symbols and colors will all have been used, making it more costly for other firms to compete. Still, there may come a time, particularly if the symbol or color in question has been used exclusively over a period of years, when the common signifier denotes the producer's brand. Not to allow exclusive use in these circumstances would destroy information capital if other producers started using the same signifier. Not surprisingly, the law allows appropriation in this case.

An interesting example is the trademarking of common shapes and colors of pills sold as prescription drugs. After a patent on a drug expires, other firms may begin selling the "same" drug under a different brand name or under its generic name while copying the shape and color of the original manufacturer's pill. Notwithstanding the lower price charged by the new entrants, many consumers may prefer to stick with the original manufacturer; maybe they had good experience with the drug and are reluctant to believe claims that the substitute is identical in all material respects. Since a consumer is unlikely to read the fine print on the pill that identifies the manufacturer (and it really is fine print), he may rely on the only accessible signifiers—its shape and color—to indicate that he is using the pill he wants. So if entrants are allowed to use the same size, shape, and color, this may lead to deliberate substitution by the druggist because the manufacturer of the generic substitute charges the druggist a lower price or because the druggist is temporarily out of the original manufacturer's drug, or to inadvertent substitution because of the druggist's carelessness. In these circumstances, where there

are large benefits from source identification and high costs of using devices other than size, shape, and color to identify brands, we would expect, and we find, that courts grant trademark protection to common sizes, shapes, and colors of prescription drugs⁴² even though they would not do this with other products. Nonprescription drugs are an example. The manufacturer can display the brand name prominently on the container and packaging, and therefore does not require size, shape, and color for source identification.

3. The Problem of Generic Names

Generic words cannot be trademarked at all; what is more, if a trademark becomes a generic name, trademark protection immediately ceases.⁴³ A generic name or term is by definition the name not of a brand but of an entire product; "airplane" and "computer" are examples of generic names. If the producer of one brand could appropriate the name of the product, he would earn rents because of the added cost to his rivals of periphrasis—of describing their products as "heavier-than-air flying machines" or "artificial-intelligence machines."

The monopoly that results from the appropriation of a generic name is conventionally described as a product monopoly but is more accurately described as a language monopoly. Unless the owner of the generic name is the lower-cost producer throughout the whole feasible range of the product's output, he will license the use of the name to competitors and receive rents in the form of license fees. Licensing will prevent firms from expending resources on developing new ways to denote their products; taking out a license is cheaper. Licensing thus transforms a social cost into a transfer payment to the firm appropriating the generic term, but it does not eliminate all deadweight losses. The licensing fee will lead to a reduced output (or higher prices) of the product, and there will also be costs of negotiating and enforcing the trademark licenses and costs of obtaining generic trademarks (including costs generated by rent-seeking behavior) in the first place. Moreover, even without licensing, although the appropriation of a generic mark will raise the costs of competing firms it need not raise them

42. See *Ciba-Geigy Corp. v. Bolar Pharmaceutical Co.*, 747 F2d 844, 224 USPQ 349 (CA 3 1984).

43. Two illustrative cases are *Miller Brewing Co. v. Falstaff Brewing Corp.*, 655 F2d 5, 211 USPQ 665 (CA 1 1981) ("Lite" beer); *Eastern Air Lines, Inc. v. New York Air Lines, Inc.*, 559 F Supp 1270, 218 USPQ 71 (SDNY 1983) ("Air Shuttle"). For a good discussion, explaining why the name "Fire Chief" on a magazine about fire fighting is not generic, see *H. Marvin Ginn Corp. v. International Assn. Fire Chiefs, Inc.*, 782 F2d 987, 228 USPQ 528 (CAFC 1986).

so high that any firm leaves the market; it might, however, reduce their competitive effectiveness.

Notice that the costs of obtaining generic trademarks (if such trademarks were allowed) would often merge with the costs of invention. The firm best situated to appropriate a generic name is the first producer of the product. Hence, if generic names could be trademarked, the rents from invention would go up because they would include rents from the name of the product. If the current legal protection of inventions is optimal (or is simply assumed to be such by courts unable to investigate the issue), any legal protection for generic names will create socially wasteful opportunities to earn rents.

All this may explain why generic names cannot be trademarked yet may seem to leave unexplained why a trademark that becomes a generic name loses trademark protection.⁴⁴ If a producer is clever enough to name his brand with a word that will some day be used as the name of the entire product, should he not be rewarded for this valuable addition to the lexicon? Is the language not richer for such words as "thermos," "aspirin," "cellophane," "dry ice," all well-known examples of trademarks that have become generic names? Our discussion of the economics of language suggests an answer. Property rights are not necessary in order to induce the rapid creation of serviceable new words for new things. Trademarks are a minor source of generic names, and we can think of no product whose introduction or diffusion was retarded because it did not have a serviceable name. Observe the rapidity with which a large vocabulary of arresting and memorable terms has emerged (without significant assistance from trademarks) to describe things that are new in the last half century—medicines, weaponry, political and social movements, legal and scientific terms, and so on. Our society seems not to suffer from such lexical poverty that a more extensive system of property rights in words is needed to alleviate it. And the trade-off between invention and access must always be kept in mind. It is well illustrated by the application of the fair-use doctrine of trademark law⁴⁵ to the controversy over the use of "Star Wars" to describe President Reagan's "Strategic Defense Initiative." STAR WARS is the trademarked name of a movie, but it has been held that the owner of the trademark cannot prevent its use to describe the Strategic Defense

44. This is an example of the fact that trademarks can be time limited, even though not subject to fixed expiration dates the way patents and copyrights are. Another example, discussed earlier, is abandonment through ceasing to sell the trademarked product.

45. See Lanham Act §33(b)(4), 15 USC §1115(b)(4); 1 McCarthy, *supra* note 1, §11.17 and 13.3; 2 McCarthy, *id.* §§31.37-38.

Initiative.⁴⁶ Such a use could not be enjoined in any event under traditional trademark law, which requires proof of at least a likelihood of consumer confusion; but were it not for fair-use considerations (that is, the high social value of allowing the trademark to be communalized), the use might be enjoined under state antidilution statutes (of which more later).⁴⁷

Cutting off trademark protection when a trademark becomes generic will reduce the costs of communication by making it cheaper for competitors of the first producer to inform the consumer that they sell the same product. It will also reduce the cost of rent seeking. And although the original producer will expend resources on preventing his trademark from becoming generic, these are not sterile expenditures; they have the unintended but important effect of reminding consumers of the existence of competing brands. Every time General Foods Corp. stresses "SANKA-brand decaffeinated coffee," it implies the existence of other brands of decaffeinated coffee. This retards the emergence of SANKA as a generic name but reinforces "decaffeinated coffee" as a generic name. There is some social cost if SANKA would be a cheaper (it certainly is a shorter) generic name, but it may be less than the cost of allowing legal protection of trademarks that become generic names.

A second cost of cutting off trademark protection when a trademark achieves generic status is that it imposes a dichotomous solution on a continuous problem. Generic status is achieved gradually. There will be a long period during which some consumers will still think of the name as the name of a particular brand and others will think of it as the name of the product. If the law waits until everyone thinks of it as the name of the product, the trademark owner will obtain substantial rents; if the law ceases to protect the trademark as soon as a few consumers think of it as the product name, this will create undue confusion and impose substantial consumer search costs.⁴⁸ In principle, trademark pro-

46. See *Lucasfilm Ltd. v. High Frontier*, 227 USPQ 967 (DC DC 1985).

47. The state dilution provisions are the following: Alabama, Ala Code §8-12-17 (Supp 1983); Arkansas, Ark Stat Ann §70-550 (1979); California, Cal Bus & Prof Code §14330 (West Supp 1984); Connecticut, Conn Gen Stat Ann §35-11(c) (West 1981); Delaware, Del Code Ann tit 6, §3313 (Supp 1982); Florida, Fla Stat Ann §495.151 (West 1972); Georgia, Ga Code §106-115 (1984); Idaho, Ida Code §48-512 (1977); Illinois, Ill Rev Stat ch 140, §22 (Supp 1984); Iowa, Ia Code Ann §548.11(2) (Supp 1984); Louisiana, La Rev Stat §223.1 (1984); Maine, Me Rev Stat Ann tit 10, §1530 (1980); Massachusetts, Mass Gen Laws Ann ch 110B, §12 (West Supp 1980); Missouri, Mo Ann Stat §417.061(1) (Vernon 1979); Montana, Mont Code Ann §30-13-334 (1983); Nebraska, Neb Rev Stat §87-122 (1981); New Hampshire, NH Rev Stat Ann §350-A:12 (Supp 1981); New Mexico, NM Stat Ann §57-3-10 (1978); New York, NY Gen Bus Law §368-d (McKinney 1984); Oregon, Ore Rev Stat §647.107 (1981); Pennsylvania, 54 Pa Cons Stat Ann §1124 (Purdon Supp 1984); Rhode Island, RI Gen Laws §6-2-12 (Supp 1983); and Tennessee, Tenn Code Ann §47-25-433 (Supp 1983).

48. This point is stressed in Folsom and Teply, *supra* note 1.

tection should cease when the costs of continued protection (dead-weight losses resulting from higher prices, which cause demand to fall; higher costs to rivals in using alternative words; and the costs of licensing and defending trademarks) exceed its benefits (diminished confusion of consumers, lower search costs, and the gains associated with the incentive to develop high quality goods). Of course the law cannot fix this point with any precision,⁴⁹ but no satisfactory alternative springs to mind; a fixed date of expiration would be wholly unsatisfactory for the reasons explained earlier.

A difficult problem of determining whether a trademark has become a generic name arises in cases in which the trademark owner initially has a product monopoly. These cases are common, not because monopolies are common but because a brand name is likely to become a generic name when there is only one brand. Facing only one producer, consumers have little incentive to use separate terms to describe the product and the brand or, what is equivalent, to denote product information and source information. It is more economical, for example, to refer to both an instant camera and its manufacturer as POLAROID than to use both terms—until Polaroid Corp. ceases to be the only manufacturer. But this also makes it difficult to distinguish between brand name and product name; the brand and the product are one. Maybe the presumption should be in favor of generic status. If the consumer has never had a competitive alternative to A's product, the brand name is less likely to convey information about the particular attributes of A's good than simply to define the product itself. Without the comparisons enabled by competition, the benefits of a trademark are difficult to evaluate because the consumer is not using A's trademark to lower his costs of searching for the brand he wants to consume.

A different point about the monopoly case will bring out the third cost of making trademark protection end when generic status is achieved. A consumer who bought Bayer aspirin at a time when Bayer was the only producer and "aspirin" was its' trademark (as it still is in Canada, Germany and certain other countries) might have assumed, when other brands came on the market and were allowed to be labeled "aspirin," that they would be identical to Bayer aspirin not only in formula but in every respect—including, for example, quality control. The difference between brand and

49. The new Trademark Clarification Act of 1984, Pub Law 98-620, 15 USC §1064(c), amending the Lanham Act, makes "the primary significance of the registered mark to the relevant public" the test of whether the mark has become generic. This formulation seems vague enough to embrace the kind of cost-benefit analysis suggested in the text.

product is not self-evident and may be difficult to determine in the period of transition from monopoly to competition. Giving generic status to the former brand name may in effect produce too much or too rapid competition for the patronage of consumers who ignore quality differences between the old and the new brands.

A suggestive or descriptive mark is more likely to become a generic name than a fanciful or arbitrary mark, though there are important exceptions, such as "aspirin" and "cellophane" ("dry ice," "thermos," and "yo-yo" are examples of a descriptive and two suggestive marks that became generic names). Why, then, would anyone choose a suggestive or descriptive mark? The answer is that, as noted earlier, such a mark conveys additional information to the consumer—to wit, information about the attributes as well as just the source of the good⁵⁰—and is thus a partial substitute for advertising. This gain has to be traded off against the increased risk of losing the mark should it become generic (as well as against its lesser distinctiveness, and, hence, lesser value as source identification compared to a fanciful or arbitrary mark). We predict that, the shorter the expected life of a brand, the more likely the producer is to use a suggestive mark. We also predict that a descriptive mark will generally be used only on brands with a long expected life, since it takes time to acquire secondary meaning, a prerequisite to obtaining trademark protection for such a mark.

Courts may not be doing a very good job of determining when trademarks have become generic, at least if the dictionary can be considered an accurate inventory of words in general use by the relevant publics (perhaps a big if). Of thirty-five illustrative examples in the McCarthy treatise of marks that courts have been held to be generic,⁵¹ sixteen either do not appear in the most recent unabridged dictionary⁵² or, if they do appear, the meaning held to be generic is not included (for example, "matchbox" is in the dictionary, but not as meaning toy cars sold in matchbox-sized boxes). Of seventeen illustrative examples of trademarks that courts have held not to be generic,⁵³ seven nevertheless are listed in the dictionary with the rejected generic meaning. Thus, although words held to be generic are more likely to show up in the dictionary than those held not to be generic, the difference in probabilities is small—fifty-four percent versus forty-one percent.

50. See John M. Carroll, *What's in a Name? An Essay in the Psychology of Reference* 179 (1985).

51. See 1 McCarthy, *supra* note 1, §12.3.

52. Random House Dictionary of the English Language (1983).

53. See 1 McCarthy, *supra* note 1, §12.4.

C. The Defense of Functionality

The concept of functionality,⁵⁴ which is mainly important in connection with design features used as trademarks (the Perrier example), is a parallel concept to genericness. A functional feature cannot be trademarked, and a trademarked feature loses trademark protection when it becomes functional. The maker of a tire could not trademark its circular shape but could trademark an irregularly shaped hubcap. The maker of a steak knife could not trademark the serrated blade but could trademark an intricate arabesque carved into the handle. A particular shape for a container might initially be subject to being trademarked, but if technological developments made it much cheaper to manufacture than alternative shapes, it would lose its trademark protection.

As the last example suggests, the concept of functionality can be given a precise economic meaning. A nonfunctional feature, hence one that can be trademarked, is one with perfect (or nearly perfect) substitutes, so that allowing a property right in the feature will create no deadweight loss. But if the feature lacks good substitutes, either because the product is worth less without it (the circular tire) or because it makes the product cheaper to produce (the example of the container shape), trademark protection will be denied. The feature may of course be a worthwhile addition to the stock of useful knowledge, but, if so, it may be patentable. Trademark protection for a functional feature would circumvent the limits, discussed briefly in Section I, on the requirements for and duration of patents.

The trickiest problem with functionality is "aesthetic" as distinct from "utilitarian" functionality. The term "utilitarian," as used in this polarity, carries its everyday sense of "practical," "down to earth," "unadorned." The illustrations given above are of utilitarian functionality in this sense. Even if a design feature merely makes a product more pleasing, however, it may be deemed functional and trademark protection will be withheld. The concept of "aesthetic" functionality gives recognition, highly appropriate from an economic standpoint, to the fact that utility in an economic sense includes anything that makes a good more valuable to consumers. But a producer of a consumer product will never deliberately uglify the product—and we do not want him to. Any design feature he seeks to trademark will be designed in part to please. Hence courts have the difficult problem of disentangling the aes-

54. On which, see, for example, Mims, *supra* note 1; A. Samuel Oddi, *The Functions of "Functionality"* in *Trademark Law*, 22 *Houston Law Review* 925 (July 1985), 76 TMR 308 (1986).

thetic from the identifying function of a trademarked design feature.

The concept of functionality, in both its "utilitarian" and its "aesthetic" aspects, can be formalized by making the cost of production of the trademarked good a function not only of the amount of the good produced but also of the physical attributes of the good that are claimed as trademarks. Assume that, if another firm is denied access to those attributes, its marginal cost of producing the good will increase; it will then be evident that exclusive rights for functional features of a product and for generic terms have the same economic effect.

A mark is not functional just because it makes the product more attractive.⁵⁵ A more attractive trademark is equivalent to a higher-quality good. So while a strong trademark increases the price that consumers are willing to pay for the good by lowering search costs, an attractive trademark raises price by increasing the utility that consumers get from the good once they have bought it. Because a more attractive mark will usually be more expensive to produce, the firm will invest in such a mark only if it is rewarded by obtaining a higher price for each unit of the trademarked good that it sells. And the higher the price, the more units the firm will produce. Thus the effect of giving the attractive trademark legal protection is to increase both the quality and quantity of output, thereby benefiting consumers, rather than to reduce output by raising rivals' costs, as in the case of utilitarian functionality.

A problem arises only if the aesthetic feature becomes an attribute of the product in the minds of consumers. In that event, to produce a brand equivalent in the consumer's mind to a brand that has this attribute a firm would have to incur additional costs, just as in the utilitarian case. The consequences for economic welfare are ambiguous. Appropriability will give the firm an incentive to spend the money necessary to produce a more attractive trademark but at the same time will raise the costs of competing firms. This is an example of true aesthetic functionality, and trademark protection is denied. Although appropriability may still be necessary to induce the expenditures required to create a pleasing design, just as to create a "utilitarian" functional design, it need not take the form of trademark protection. The law authorizes the grant of "design patents" for up to fourteen years to reward the investors of pleasing (but not "utilitarian") designs.⁵⁶

55. See, for example, *LeSportsac, Inc. v. K Mart Corp.*, 754 F2d 71, 76-78, 225 USPQ 654 (CA 2 1985).

56. See 35 USC §§171-73.

It might seem that if a design feature is both functional and identifying, the law should trade off the benefits of a more attractive trademark, which will fall if trademark protection is denied, against the costs of other producers, which will rise if trademark protection is allowed. But instead the law denies protection automatically once the feature is adjudged functional. This makes economic sense. A functional feature is unlikely to identify a particular brand, that is, to serve a legitimate trademarking (brand-identifying) purpose. By definition it is a feature that the consumer associates with the product rather than the brand (for example, the oval shape of a football). As a feature becomes functional, the producer will place increasing emphasis on the brand name, rather than the feature, to identify the brand. The tricky problem is to determine when the feature has become functional, that is, when its value in making the product more pleasing, or otherwise more valuable, or simply cheaper to produce, overtakes its value in preventing consumer confusion.

D. Infringement and Confusion

1. Why Require Proof of Actual or Even of Likely Confusion?

The ownership of a valid trademark does not carry with it as broad a right to exclude others from using it as the ownership of a piece of land does, especially in jurisdictions where dilution (discussed in the next subsection) is not actionable. In order to prevent another seller from selling his goods under your trademark, you must show that consumers are likely to think that it is your brand that he is selling.⁵⁷ This is a sensible restriction on the scope of the property right if the function of a trademark is, as we have been assuming, to name a brand. Just as people in different parts of the country or in different occupations can have identical names without causing misunderstanding, so sellers in unrelated product or geographic markets should be able to affix the same names or other marks to their goods without confusing consumers. Suppose A and B produce different brands of product X. A has a strong mark (implying low search costs for consumers). B adopts a similar mark, but there is no likelihood of confusion between the two marks. Then both marks will convey accurate information about the reputation of the underlying product (or

57. This is a lesser burden than having to prove actual confusion. The choice between a "likelihood" and an "actual" standard involves balancing two potential error costs. Under the former standard, some similar marks that do not cause confusion will be held to be infringing. Under the latter, some similar marks that do cause confusion will be held to be noninfringing. We do not analyze the choice between the two standards.

producer). A's brand will command a higher price than B's because the search costs associated with the former are lower, but this is consistent with competition and with maximizing the sum of consumer and producer surplus. There will be no free riding, because B will be unable to appropriate any of A's return from A's high reputation for quality. B's revenue depends on the search costs associated with its, not with A's, brand, the absence of confusion implying that consumers correctly match search costs with the brand.

Legal intervention to prevent B from using a mark that is similar to A's but unlikely to cause confusion would impose litigation and other costs (for example, the cost of B's changing its trademark) without any offsetting benefits. The costs could be heavy. A seller might adopt a trademark in all innocence, not knowing that some other seller, selling a different product in a remote area of the country, had adopted the same trademark previously; he might invest substantial resources in advertising his trademarked good; and he might be forced to write off the entire investment if first use established a nationwide property right covering all products. To avoid such disasters, sellers would have to invest heavily in investigating prior uses of trademarks they were thinking of adopting. These costs are reduced if the original owner must show a likelihood of confusion.

It is consistent with this analysis that the owner of a registered mark bears a lighter burden of proving likelihood of confusion, as we saw earlier. Registration warns off potential infringers in a way that mere use does not; in the example just given, the infringer, but for the trademark registry, might have no idea that the same product was being sold under the same name elsewhere in the country.

2. Determining Likelihood of Confusion

In deciding whether there is a likelihood of confusion, courts look at such things as the similarity of the original and allegedly infringing trademark (for example, EXXON and EXXENE), the strength of the original mark, the similarity of the products involved, whether the consumers of the products overlap, whether the products are sold through the same retail outlets, and how knowledgeable the consumers are.⁵⁸ The last point is particularly interesting from an economic standpoint, because it shows that the inputs into the sale of a product include information supplied

58. See, for example, *In re E.I. duPont De Nemours & Co.*, 476 F2d 1357, 1361, 177 USPQ 563 (CCPA 1973); 1 Gilson, *supra* note 1, §5.01.

by buyers as well as by sellers. The cheaper the information produced by buyers is—perhaps because the consumers of a particular product are particularly knowledgeable, such as business purchasers of a supply essential to their business—the less information the seller must supply as by making a great effort to distinguish his trademark from a competitor's.

To formalize the analysis just a bit, notice first that a consumer confronted by one of two confusingly similar marks (those of A's and of B's brands) will be uncertain whether the search costs associated with the trademarked item are those associated with A's trademark or those (assumed-to-be-higher search costs) associated with B's. Facing A's brand in fact, but adjusting for the possibility that it may really be B's rather than A's, the consumer will pay less than he would if he were sure it was A's. The price of each unit of A will fall, and the fall will be greater the stronger A's trademark is relative to B's and the greater the likelihood of confusion. At the same time, consumers will underestimate the search costs associated with B's product because they will attach a positive probability to B's product actually being A's. This will lead to a higher price for B's product—higher the greater the likelihood of confusion and the stronger A's trademark relative to B's.

The net effect on consumers of B's adopting a mark confusingly similar to A's will at first be zero. If the likelihood of incorrectly identifying A's and B's goods is identical, consumers paying a lower price for A's good will just offset those paying a higher price for B's, though a deadweight loss will result because less of A's good and more of B's will be produced than consumers really want. But confusion between the two marks will lower A's profits and thus ultimately harm consumers even more. For A will respond to the confusion by reducing both his output and his trademark expenditures (since those expenditures now yield less benefit). The reduction in output will lower consumers' expectations that the brand they are buying will really be A's rather than B's and thus force A to reduce his price still further—leading to a further reduction in both his output and his trademark investment. This process may continue until eventually the information conveyed by A's trademark is little more than that conveyed by B's. Thus B's infringement will have harmed consumers by eliminating A's incentive to maintain a stronger trademark than B's and hence produce a more valuable product (one with lower search costs).

We have been assuming that firms produce identical physical products but different brands and that A's brand has the better reputation, but the assumption is unrealistic. When B adopts a mark similar to A's for the purpose of confusing consumers, B is also likely to produce a lower-quality product than A. Consumers who confuse B's mark with A's will assume that the quality of the underlying physical good is the same; and if B can cut costs by cutting product quality without consumers' becoming aware, he will have even higher profits than he would by maintaining the same quality as A.⁵⁹ Moreover, B's incentive to free ride on A's trademark will be greater the higher the quality of the underlying good (adjusted for B's costs of making the physical good appear equivalent to A's). Without trademark protection, therefore, A would have less incentive either to develop a strong trademark or to produce a high-quality good.

The factors that courts use to estimate the likelihood of confusion bear directly on A's losses, and, ultimately, therefore, on A's incentive to adopt a stronger mark. Similarities in the appearance and sound of the two marks, buyers' lack of sophistication, similarity of the underlying product, and overlapping sales territories all raise the probability of confusion; and the stronger A's trademark is, the greater will be the price reduction for A's good if infringement is not prevented. The same factors also affect B's gain from free riding. Notice in particular that B has an incentive to adopt a mark similar to A's only if A has a better reputation than B, so that the search costs that the buyer must incur to be sure of obtaining satisfactory quality are lower in buying from A than from B. Only then can B charge a higher price for a unit of X and earn higher profits, assuming that the physical costs of duplicating A's trademark are not so great as to make infringement unprofitable. It would be surprising, therefore, if firms intentionally infringed weaker trademarks.⁶⁰

59. In the previous discussion, we assumed that A and B produced items having the same expected physical characteristics, but that B's brand was of lower quality than A's because, for example, the reputation embodied in its trademark was poorer as a result of a greater variance in the quality of the brand, requiring consumers to search more to be sure of getting the quality they want. Because search costs are greater, consumers are unwilling to pay as high a price for B's good.

60. The analysis is more complicated, although the conclusion is the same, if the quality of the underlying good differs among firms. A firm producing a high-quality-weak-trademark good might infringe the trademark of a low-quality-strong-trademark good because the gain in consumer recognition would more than offset the reduced consumer perception of the quality of the underlying product. *Taylor Wine Co. v. Bully Hill Vineyards, Inc.*, 569 F2d 731, 733-34, 196 USPQ 593 (CA 2 1978), modified 201 USPQ 65 (CA 2 1978), was such a case. The defendant used the well-known "Taylor" name in its trademark, although it made higher-quality wines than the Taylor wines.

3. How Likely Must Confusion Be? Herein of Intent

A final question about confusion is, how many or what proportion of consumers must be confused (or are likely to be confused) before infringement will be found? Consumers differ in their ability to distinguish among trademarks, according to how careful they are in searching for goods and to their intelligence.⁶¹ Suppose B adopts a trademark that is similar but not identical to that of a similar product sold by A. Careful consumers, defined as consumers with low information costs, are not fooled, but careless consumers—defined not as consumers who do not care about brands but as consumers with high information costs (equivalent in the economics of tort law to potential accident victims who have above-average costs of taking care)⁶²—are confused. Craswell is concerned with the situation in which removing all ambiguity from an advertising claim for the sake of the careless may make the advertisement confusing for the careful (or, perhaps, for some other group of careless who, however, were not deceived by the original claim). That problem is less acute in the trademark context. B should be able to find a trademark that distinguishes his product from A's in the minds of the careless without confusing the careful (or other careless). We would therefore predict that courts would be more protective of the careless consumer in the trademark setting than in false advertising cases.

We would also predict that the allegedly infringing use would be enjoined if the plaintiff (A in our example) could show that B had adopted its similar trademark with intent to deceive, even if only the most careless consumers would be deceived.⁶³ This is a case in which the cost of preventing confusion is negative, so that, even if the benefits are slight, prevention is cost-justified unless the costs of using the legal system are very high.⁶⁴ The case in which the second seller incurs costs to devise a confusingly similar trademark is thus fundamentally different from the case in which the second seller would have to incur (modest) costs to avoid confusion. Furthermore, the more the infringer spends on duplicating another firm's mark, thereby creating the impression that there is a single source of these two goods, the greater will be the number

61. Compare Richard Craswell, *Interpreting Deceptive Advertising*, 65 *Boston University Law Review* 657, 672-84 (July 1985).

62. See, for example, Landes and Posner, *supra* note 3, ch 10.

63. See, for example, *American Chicle Co. v. Topps Chewing Gum, Inc.*, 208 F2d 560, 563, 99 USPQ 362 (CA 2 1953) (L. Hand, J.); compare *My-T Fine Corp. v. Samuels*, 69 F2d 76, 21 USPQ 94 (CA 2 1934) (L. Hand, J.); 2 McCarthy, *supra* note 1, §§23.30 and 32.

64. This is the economic definition of an intentional tort for which no defense of, or akin to, contributory negligence (that is, no defense of victim fault) should lie. See Landes and Posner, *supra* note 3, ch 6.

of consumers who are likely to be confused (given the underlying distribution of abilities and care among consumers).

Now suppose that the infringement is unintentional. B adopts a mark not knowing that it is similar to A's and likely to confuse consumers. Maybe B was unaware of A's mark because the geographic overlap between A's and B's markets is small or because their products appeal to different consumers. The analysis suggests that the question of how many consumers must be confused before a court finds that B has infringed A's mark should be decided by balancing costs and benefits. B's trademark provides benefits to those consumers who are not confused and who use B's mark to identify and distinguish B's good, thus lowering their search costs and increasing B's output. But it harms those consumers who believe that they are getting a good produced by A. If the additional cost to B of reducing confusion is greater than the expected reduction in harm, B should not be found to have infringed. If the situation is reversed, it should be.

The courts do not balance costs and benefits explicitly in deciding whether there has been infringement, but such a calculus may be implicit in the judicial requirement that in a case of non-deliberate infringement the plaintiff must prove that an appreciable number of ordinarily prudent consumers are likely to be misled by the similarity between the two marks.⁶⁵ The more who are misled, the likelier it is that the costs of confusion will exceed the costs of changing the second mark. If the law found infringement when only a trivial number of careless consumers were likely to be confused, the benefits in reduced confusion would probably be less than the costs in making the owner of the second mark adopt a new and even more dissimilar mark (or have adopted such a mark before beginning to sell this product). But if the law required that all or the vast majority of consumers be confused before infringement could be found, the balance would tip too far the other way, for the cost of finding a new mark (especially *ex ante*) is not so great that it is likely to exceed the cost of a large number of consumers' being confused. Notice how the focus on the ordinarily prudent consumer allows for the possibility that the lowest-cost avoiders of confusion may be the consumers themselves, in which event the burden of avoiding confusion is placed on consumers, by refusing to enjoin the second mark.

It is interesting to compare the approach taken by trademark law to the problem of consumer confusion with that taken by the

65. See, for example, *McGregor-Doniger, Inc. v. Drizzle, Inc.*, 599 F2d 1126, 202 USPQ 81 (CA 2 1979); 2 *McCarthy*, *supra* note 1, §23.1(B).

Federal Trade Commission to the parallel problem of misrepresentations about quality. The Commission's efforts in this area have invited a drumbeat of criticism,⁶⁶ which the common law of trademark infringement has escaped. In part this is due to doctrinal differences; the Commission protects (or, at least, purports to protect) careless consumers from even innocent misrepresentations,⁶⁷ while trademark law, as we have seen, protects the careless consumer only from deliberate misrepresentation. In part it may be due to the fact that, since a firm that complains to the Commission bears none of the costs of enforcement (they are borne by the taxpayer if the Commission decides to act on the complaint), it has less incentive to avoid making frivolous and anticompetitive complaints than it would have if it bore part of those costs, which it would do in a private lawsuit.⁶⁸ Even more important is the difference between common law and public regulation as methods of promoting the efficient use of resources. Studies have shown that common law fields (which trademark law mainly is, despite the Lanham Act) are more likely to be influenced by a concern with achieving efficient resource allocation than administrative regulation is, for a variety of institutional reasons.⁶⁹ This study provides further evidence of the difference.

E. Dilution

Suppose a lounge in Boston calls itself "Tiffany's" or a peanut vendor in the Bowery calls himself "Rolls-Royce Ltd." There is no danger that consumers will think that they are dealing with Tiffany's or Rolls-Royce if they patronize these sellers, so it might seem that there would be no case for thinking them guilty of trademark infringement. Many states, however, would recognize a cause of action in such a case today for "dilution" of the Tiffany's and Rolls-Royce trademarks.⁷⁰ A related problem—where, however, a cause of action is not recognized—is that of cheap copies,

66. See, for example, Richard S. Higgins and Fred S. McChesney, Truth and Consequences: The Federal Trade Commission's Ad Substantiation Program, 6 *International Review of Law and Economics* 151 (December, 1986); Richard A. Posner, Regulation of Advertising by the FTC (Evaluative Study No. 11, American Enterprise Institute, November 1973); and references cited in both these works.

67. See Craswell, *supra* note 61 at 697.

68. And if the suit is deemed frivolous, the plaintiff will have to pay the defendant's attorney's fees as well as his own. See, for example, *Blau Plumbing, Inc. v. S.O.S. Fix-It, Inc.*, 781 F2d 604, 612, 228 USPQ 519 (CA 7 1986).

69. See Posner, *supra* note 4, §12.9 and ch 19; Landes and Posner, *supra* note 3.

70. See, for example, *Hyatt Corp. v. Hyatt Legal Services*, 736 F2d 1153, 1157-59, 222 USPQ 669 (CA 7 1984); 2 McCarthy, *supra* note 1, §24.13.

as where a perfume manufacturer advertises a very cheap perfume as a copy of Chanel No. 5.⁷¹

There are three possible economic grounds for the extension of the property right in the trademark beyond its use to identify a brand. The first is that there is a potential for confusion: hereafter when the consumer sees the name "Rolls-Royce" he will think about both the auto manufacturer and the peanut vendor, and the connotations of the name will blur. The problem is akin to that which leads some people to change their name if a namesake becomes notorious. There are very few people any more named "Hitler," not so much because the name would cause confusion as because it has distracting and inappropriate associations. So the communicative value of "Rolls-Royce" and "Chanel No. 5" is diluted in the cases we have put, although in the second case there may be an offsetting benefit: potentially valuable information is conveyed, and circumlocution would be costly. This leads us to predict that the cheap copy will be less likely to run afoul of antidilution statutes than would appropriation of a well-known trademark for an unrelated product.

The second economic reason for antidilution laws is based on external benefits. Rolls-Royce and Chanel Inc. have made substantial investments in creating prestigious names, and the peanut vendor or (less clearly) the cheap copier appropriates some of the benefits of those investments without compensating the investor. If appropriation is forbidden without the trademark owner's permission, the benefits will be internalized and the amount of investing in creating a prestigious name will rise. Those who believe that "product differentiation" is a bad thing because it creates artificial barriers to entry will not applaud such a result, but as we noted earlier this view is no longer widespread among economists. The economic objection to the argument from external benefits lies elsewhere. Since the number of prestigious names is vast (and, as important, would be vast even if there were no antidilution laws), it is hard to see how any of their owners could obtain substantial license fees. Competition would drive the fees to zero, since

71. See *Smith v. Chanel, Inc.*, 402 F2d 562, 159 USPQ 388 (CA 9 1968); 2 McCarthy, *supra* note 1, §25.14. A legal-doctrinal reason for this result is that the copier is not using the trademark to denote his cheap copy, just as we are not using "Chanel No. 5" to denote any product of ours merely by using the name in this article. In the Chanel case, the copier is using "Chanel No. 5" to inform consumers about the scent of his perfume. Moreover, it would be very costly for consumers to acquire such information before purchasing the copier's perfume because the perfume was sold through the mail. On remand in *Smith v. Chanel, Inc.*, the court found that the copier had violated section 43(a) of the Lanham Act (the section dealing with false advertising) because its perfume did not have the identical scent as Chanel No. 5, as claimed in its advertising. See *Chanel, Inc. v. Smith*, 528 F2d 284 (CA 9 1976).

if a name is being used in an unrelated market, virtually every prestigious name is a substitute for every other. The peanut vendor could call himself "Tiffany's" just as well as "Rolls-Royce"; he is not confined in searching for names to a particular product market. The external benefits, in short, seem slight.

The last economic ground for trademark protection on the basis of dilution relates to cheap copies and is suggested by Higgins and Rubin, who note that "many persons purchase branded goods for the purpose of demonstrating to others that they are consumers of the particular goods"⁷²—in other words, to impress others. Higgins and Rubin treat this as a pure consumption preference, but we prefer to view it as investment in reputation capital. Just as people conceal their undesirable characteristics in order to create or protect such capital,⁷³ so they flaunt their desirable characteristics. They advertise themselves (much as sellers of goods advertise their goods) by wearing clothes, jewelry, or accessories that tell the world that they are people of refined (or flamboyant) taste or high income. If others can buy and wear cheap copies, the "signal" given out by the purchasers of the originals is blurred.⁷⁴ The perfume you smell may be Chanel No. 5, which tells you something about the wearer, or it may be some cheap copy. It may be difficult or impossible to tell which.

The twist here is that the confusion does not occur in the market for the trademarked good, or in any other product market, but in a "resale" market where consumers of the product compete with other consumers for advantageous personal transactions. Using trademark law to make it harder to market cheap copies (say, by forbidding the maker of the cheap copy of Chanel No. 5 to mention Chanel in its advertising) promotes competition in this market while impairing it, perhaps severely, in the product market. The trade-off would be simple only if we were confident that the sole motive for buying the cheap copy was to pass oneself off as having a higher income; then one could regard the seller of the cheap copy as a kind of contributory infringer whose sole commercial function was to make it easier for consumers to deceive the people with whom they transact in the market for personal

72. See Richard S. Higgins and Paul H. Rubin, *Counterfeit Goods*, 29 *Journal of Law and Economics* 211 (October, 1986).

73. See Posner-Economics, *supra* note 15 at 232-42.

74. This was the basis for sporadic efforts to regulate luxury in dress in medieval Europe. In the fourteenth century, "nothing was more resented by the hereditary nobles than the imitation of their clothes and manners by the upstarts. . . . Magnificence in clothes was considered a prerogative of nobles, who should be identifiable by modes of dress forbidden to others. . . . [S]umptuary laws were repeatedly announced, attempting to fix what kinds of clothes people might wear." Barbara W. Tuchman, *A Distant Mirror: The Calamitous 14th Century* 19 (1978).

relations and, sometimes, in the job market. If this is not the sole motive, the effect of allowing damages for dilution may be to prevent the marketing of imitations, resulting in higher prices because of reduced competition. Suppose someone really could duplicate the scent of Chanel No. 5; how could it describe its product accurately without mentioning the Chanel brand? Trade-mark protection here would have the same effects as allowing a descriptive mark to be trademarked without proof of secondary meaning.
